

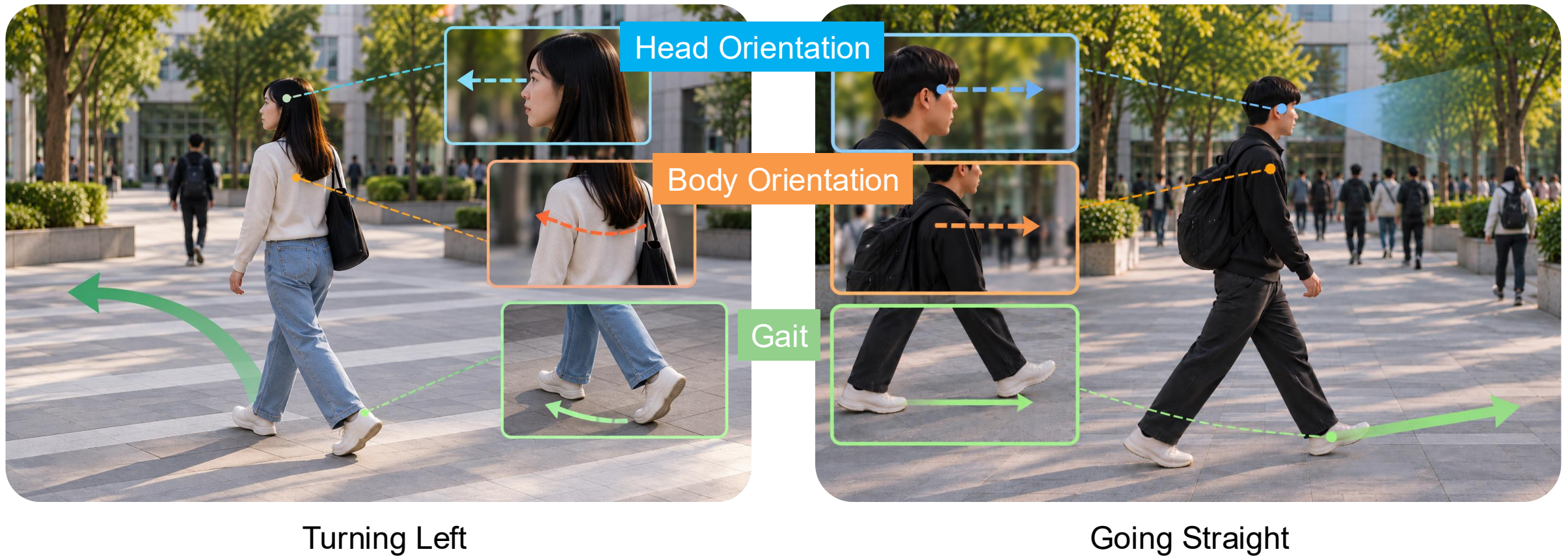
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HumAIN: Human-Aware Implicit Social Robot Navigation

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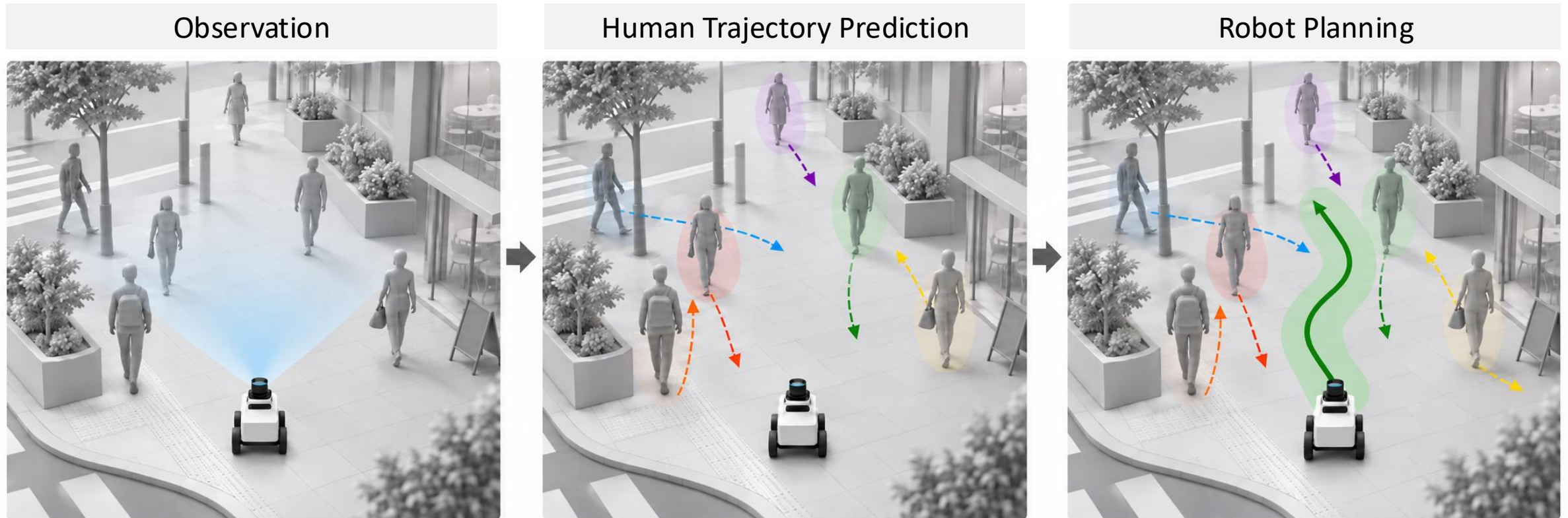
Motivation

- Humans reveal their navigational intent through **rich implicit social cues**.



Motivation

- In existing approaches, human motion is typically modeled **explicitly** before planning robot actions.



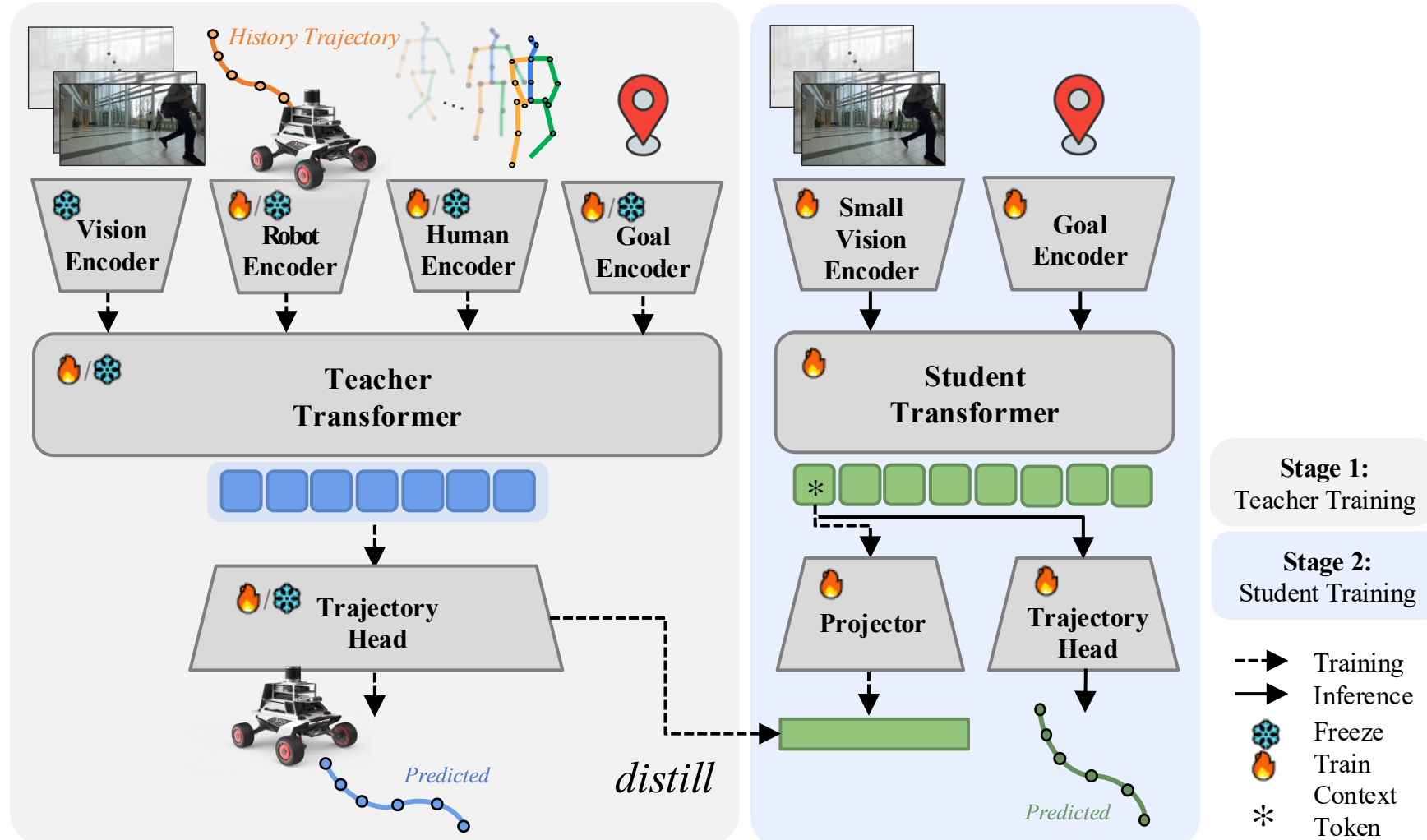
Problem: prediction errors propagate to planning.



Can we teach a robot to plan a socially compliant trajectory, by learning implicit social cues, *implicitly*?

System Architecture

- Two-stage **teacher-student learning** to distill privileged social representations





X2



Ours

Clearly avoids a person



ViNT

Bumps into a person



NoMaD

Mostly just goes straight



GNM

Results

Offline Evaluation on SCAND and OOD environments



HumAIN consistently achieves lower trajectory errors against ground-truth trajectories across diverse environments.

■ Takeaway

- **HumAIN** enables human-aware navigation through **implicit social reasoning**.



Learn **explicitly** during training. Reason **implicitly** during deployment.

Thank You!

See you at **Pod C10**
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